**25901102** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCBTS200 - GENETIC ENGINEERING AND PLANT TISSUE CULTURE

(2024 ADMISSION ONWARDS)

**Duration: 1 hour 30 minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. The enzyme used in PCR is _____ DNA polymerase. [K] / [CO2]
2. Hybridization is based on complementary _____ pairing. [K] / [CO2]
3. _____ is the optimal pH of plant tissue culture media. [K] / [CO5]
4. _____ proteins help plants from temperature stress. [K] / [CO4]
5. _____ is an aplication of ovary culture. [K] / [CO7]
6. Terminal transferase adds nucleotides to the _____ end of DNA. [K] / [CO1]
7. Monnier developed a method for the development of _____. [K] / [CO7]
8. The first commercialized Bt crop was _____. [K] / [CO4]
9. The repair mechanism that causes insertions or deletions is _____. [K] / [CO4]
10. The plasmid vector pBR322 was constructed by _____ and Bolivar. [K] / [CO1]

**[1x10 = 10]**

### Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Mention two applications of cDNA libraries. [K] / [CO2]

12. List any two applications of antisense therapy. [K] / [CO4]

13. What is the significance of somatic embryogenesis? [K] / [CO7]

14. Give two examples for somatic hybrid. [K] / [CO8]

15. What is lipofection? [K] / [CO3]

16. Name the Indian scientists who pioneered the technique of triploid production. [K] / [CO7]

17. Mention any two methods used for quantification of DNA and RNA [K] / [CO2]

18. Explain electrofusion of protoplasts. [K] / [CO8]

19. What is herbicide resistance in transgenic plants? [K] / [CO4]

20. Describe one application of single cell clones in biotechnology. [K] / [CO7]

21. Why is RNA more difficult to isolate than DNA? [U] / [CO2]

22. Differentiate between kinase and phosphatase [U] / [CO1]

**[2x10 = 20]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Describe the steps involved in micropropagation. [K] / [CO5]

24. Discuss the advantages and limitations of Sanger sequencing. [U] / [CO2]

25. Explain the differences in detection methods used in Southern and Western blotting. [U] / [CO2]

26. Discuss the scope of genetic engineering in agriculture. [U] / [CO1]

27. Explain the types of gene therapy. [U] / [CO4]

28. Describe the method of cryopresevation. [K] / [CO8]

29. Wht is callus culture? What are the applications of callus culture? [K] / [CO7]

**[4x5 = 20]**